Definition (Semitraced strict monoidal semicategory)
A symmetric strict monoidal stacking semicategory $\langle \mathbf{C}, \otimes, \mathbf{1} \rangle$ is called *semitraced* if it is equipped with a family of semitrace functions

$$\mathrm{Tr}^{Z}_{X,Y} : \mathrm{Hom}_{\mathbf{C}}(X \otimes Z; Y \otimes Z) \to \mathrm{Hom}_{\mathbf{C}}(X; Y),$$

satisfying the axioms for a symmetric stacking semicategory, plus the following axiom:

1. *Compatibility with the monoidal unit:*

   For any morphism $f : X \to Y$ in $\mathbf{C}$,

$$\mathrm{Tr}^{\mathbf{1}}_{X,Y}(f) = f.$$